

Experiment 1: where does synchrony end?

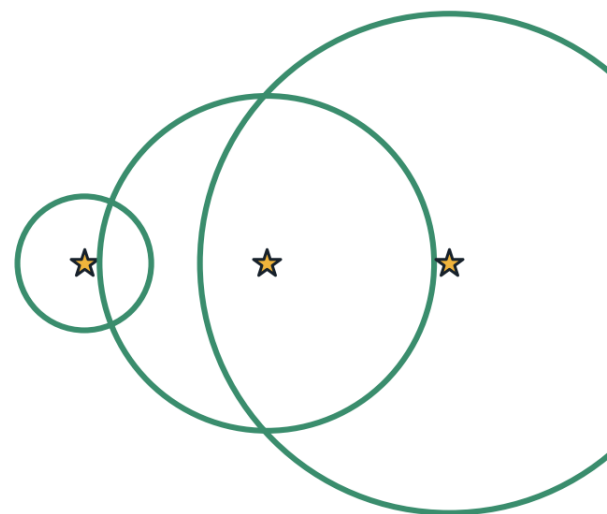
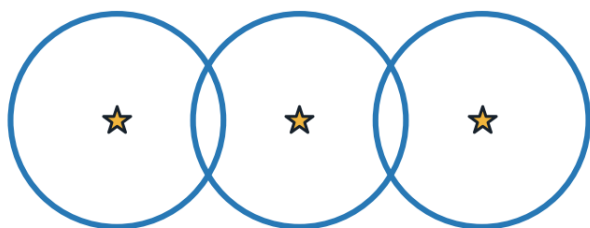
1. ORIGINAL QUESTION

The earlier experiment asked whether cold and warm synchrony reached a stable distant background within 500 km. That finite-horizon question remains preserved as a valid negative result.

The experiment changes neighbor selection only

Fixed: every focal pixel gets 100 km

Question: does empirical support differ?

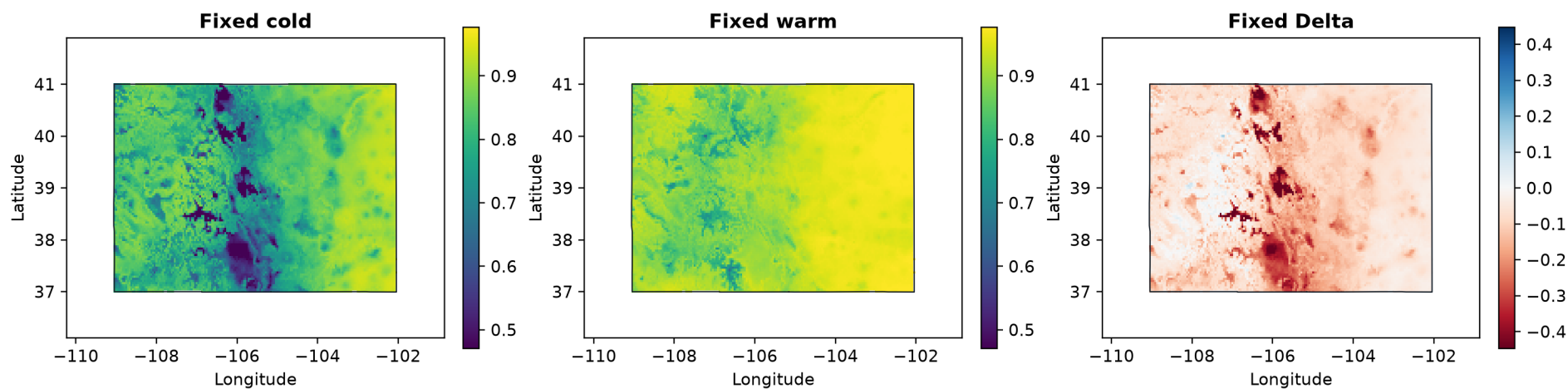


The fixed 100 km baseline remains unchanged

2. VALIDATED CONTROL

The fixed-radius Colorado products retain the audited cold lower-tail TMIN, warm strict upper-tail TMAX, and pairwise Delta_S = S_cold - S_warm semantics. Experiment 2 changes none of these products.

Validated full-state 100 km control remains unchanged

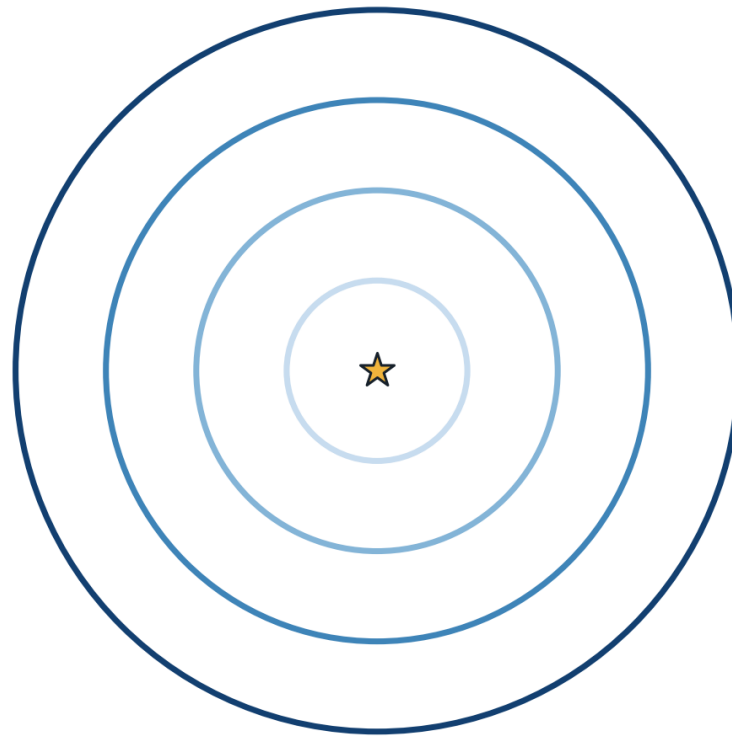


Why we tested a finite R^*

3. EXPERIMENT 1 RATIONALE

A focal-specific horizon could in principle distinguish local structure from distant background. The discovery radius was observation support, never an estimate, and unresolved cases were kept missing.

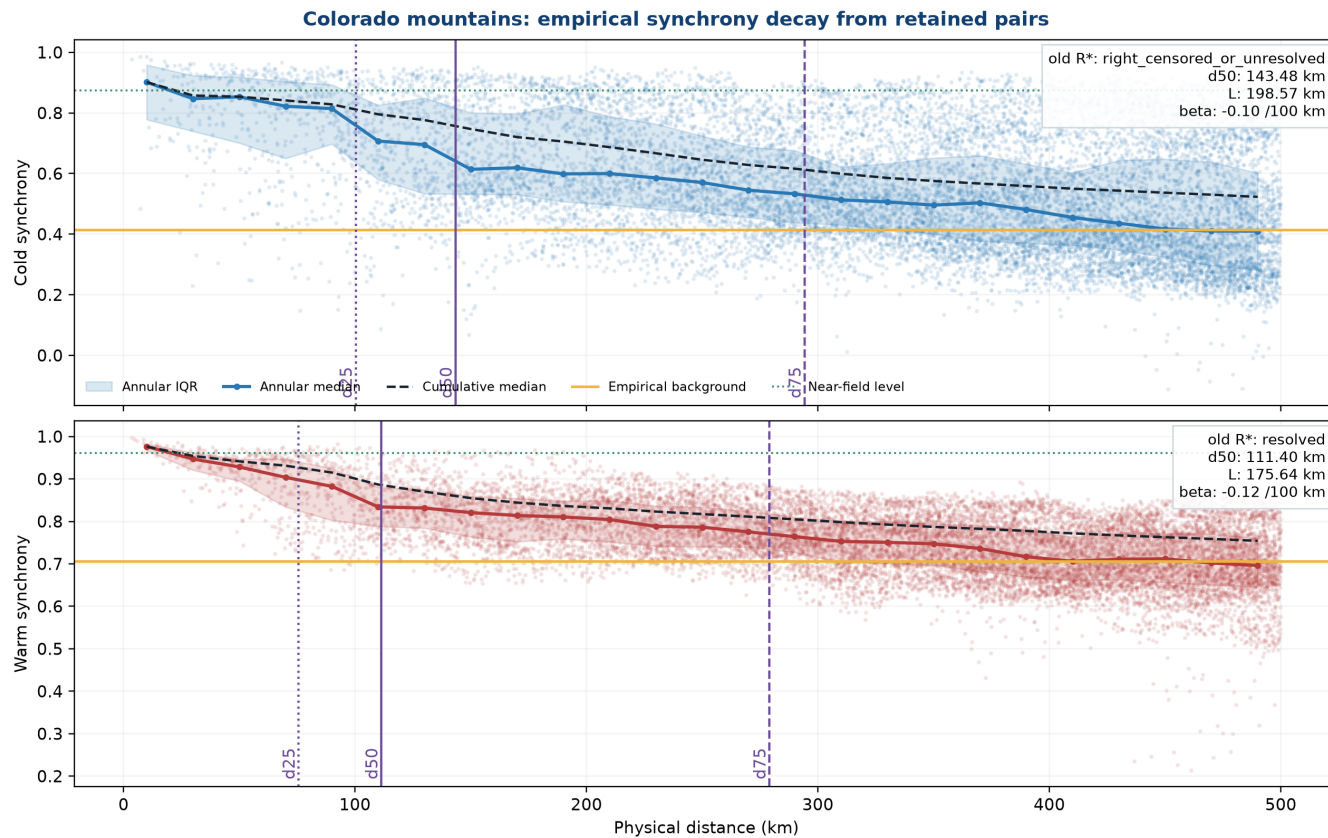
One focal pixel inside large discovery support



The empirical distance curves are real

4. RETAINED EVIDENCE

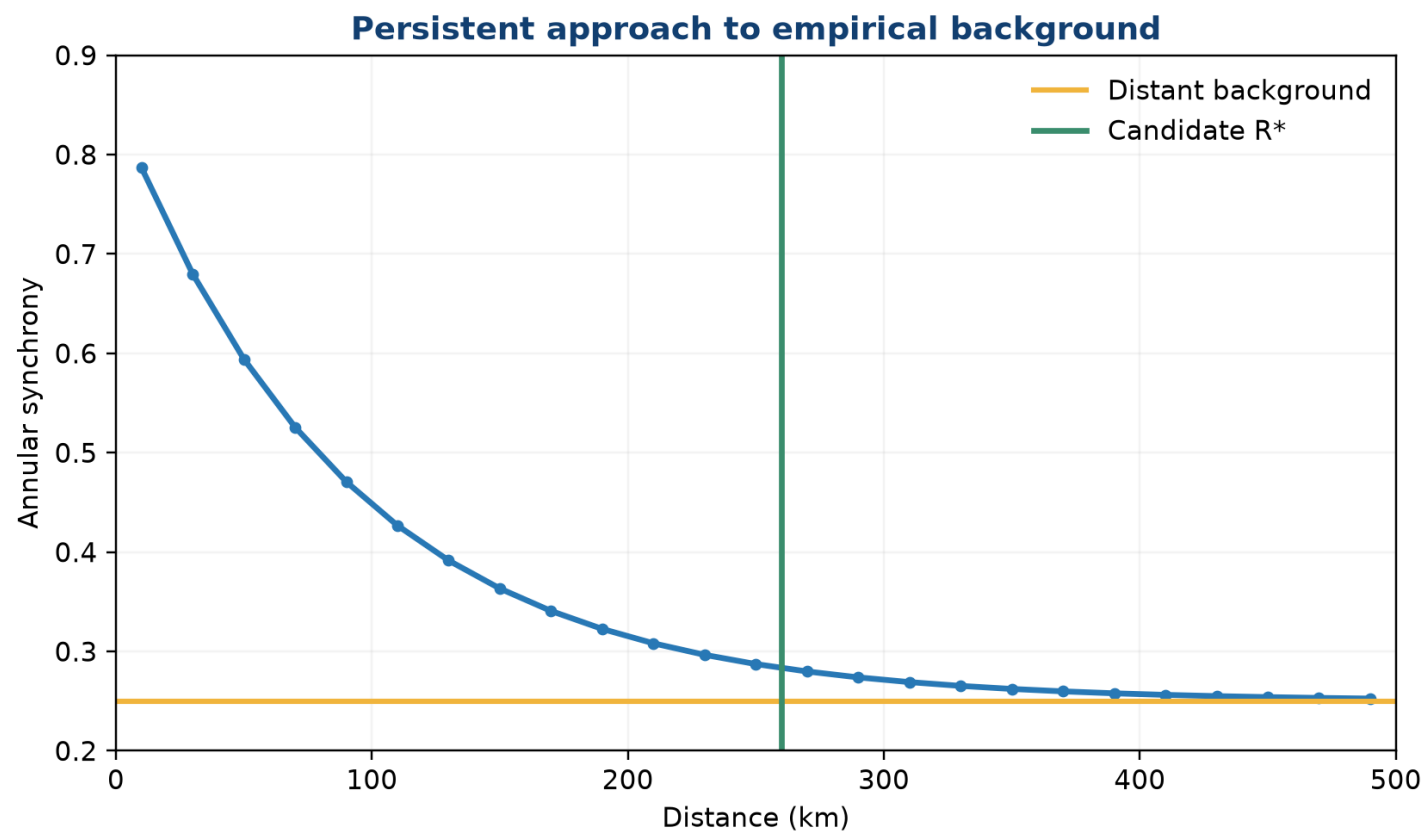
Individual pair values, annular median and IQR, and cumulative median show reproducible spatial structure. The curve is not forced to be monotonic and is not fitted to a parametric kernel.



The hard-horizon criterion

5. EXPERIMENT 1 RULE

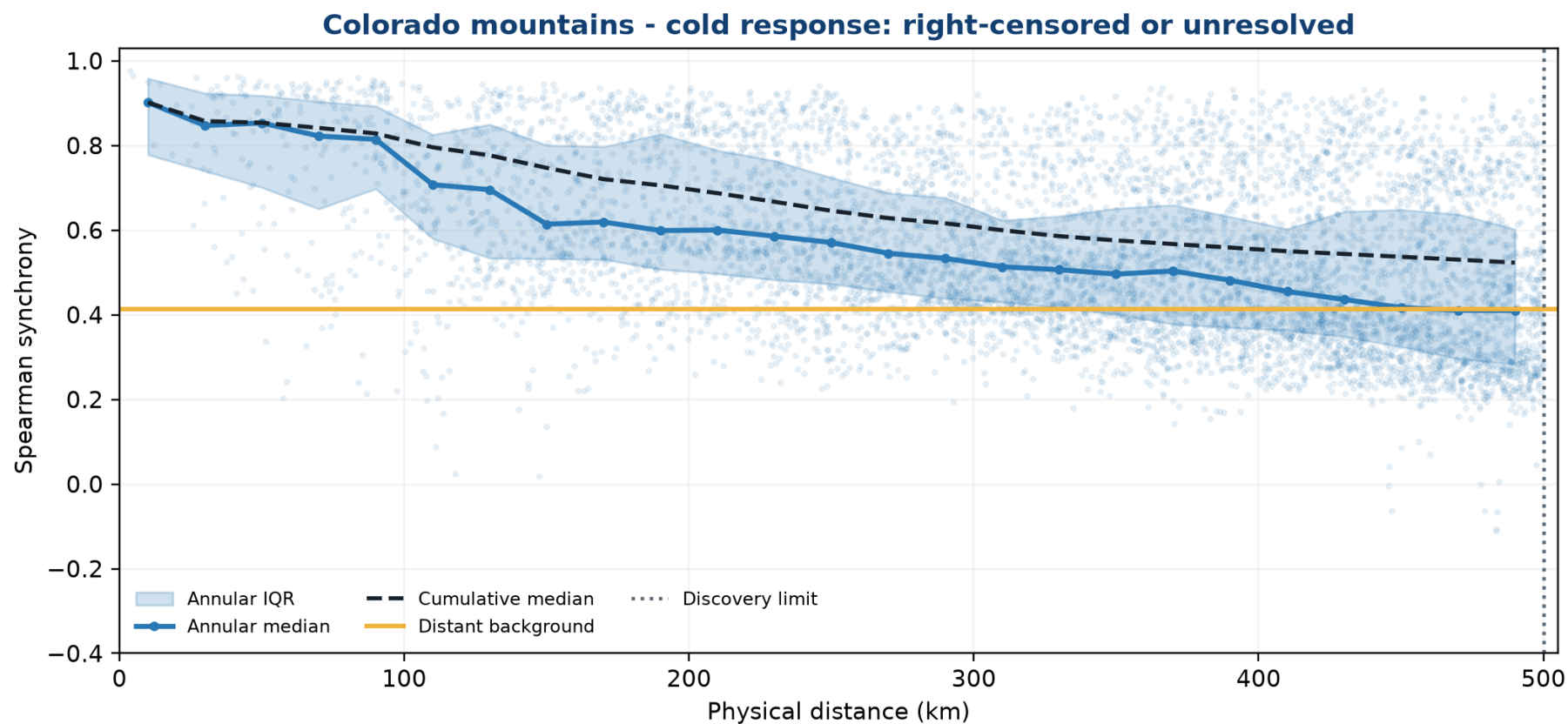
R^* required a persistent approach to an empirical nonzero background plus cumulative stability and later-shell agreement. Weakening the rule after seeing failures would invalidate the gate.



Why the horizon often failed

6. BOUNDARY AND LONG TAILS

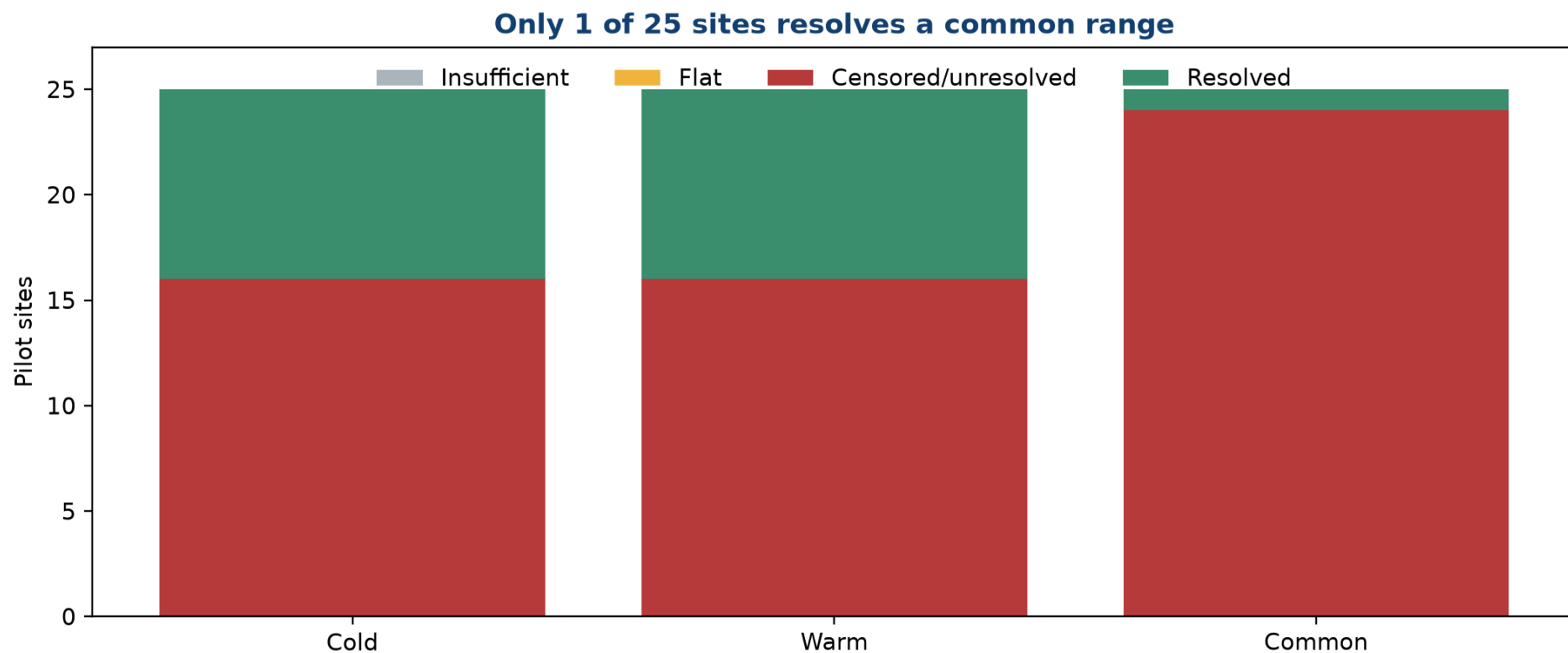
Real curves commonly continued changing or failed the later-shell safeguard. A discovery limit cannot be assigned as R^* : doing so would turn censoring into a false finite estimate.



Only 1 of 25 common Colorado ranges resolved

7. EXPERIMENT 1 RESULT

Cold and warm each resolved at nine pilot sites, but both resolved together at only one. The common-radius requirement therefore removed 96% of pilot sites.

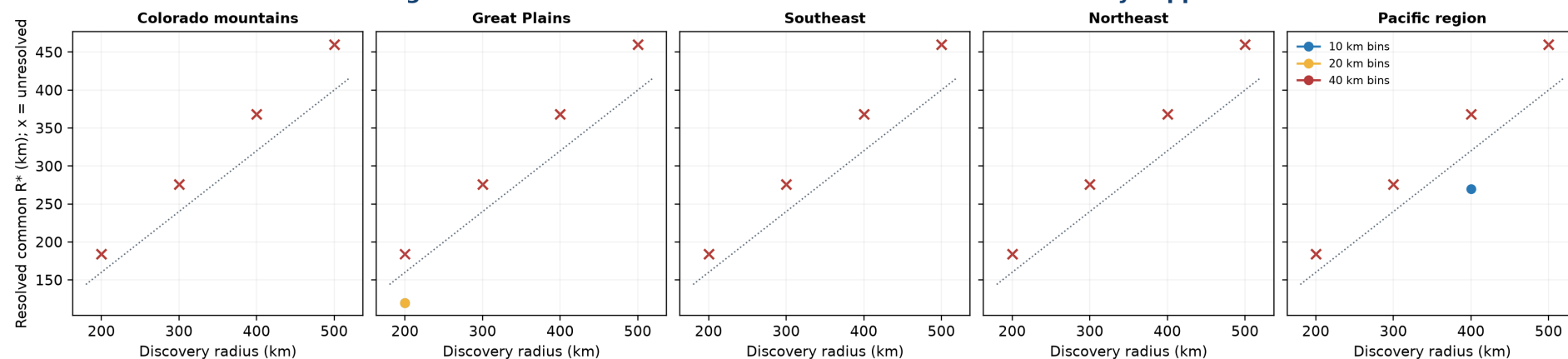


No site stabilized from 400 to 500 km

8. DISCOVERY SENSITIVITY

At the primary 20 km bin width, zero of 25 Colorado common ranges stabilized between 400 and 500 km. Five representative national pixels also yielded zero common R^* values.

Range estimates do not stabilize across bin width and discovery support



A negative horizon result is scientific information

9. INTERPRETATION

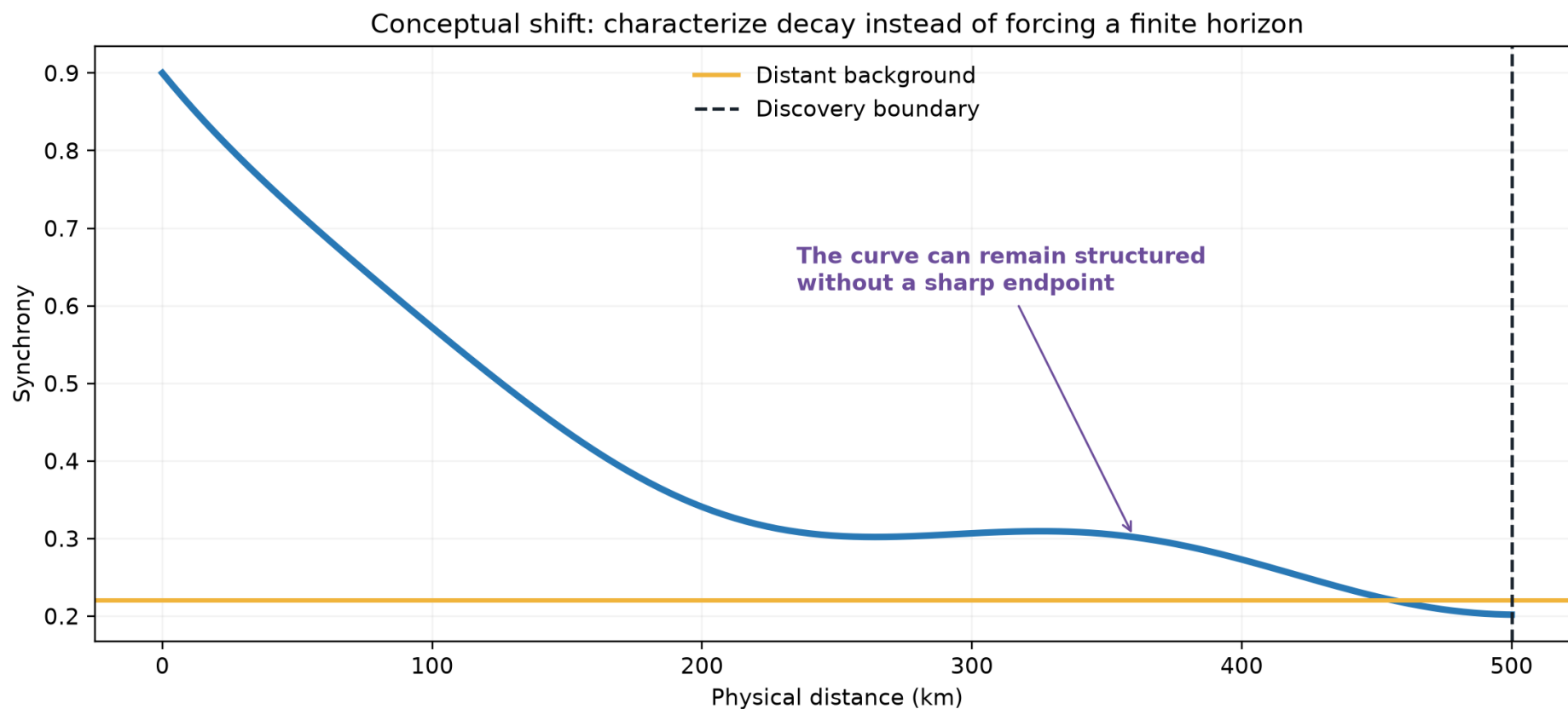
The failed gate does not imply an absence of spatial decay. It shows that these curves do not generally support a sharp, finite endpoint within the observed 500 km domain.

Observed	Meaning
0 of 5 representative common R*	No robust finite horizon at the national test pixels
1 of 25 Colorado common R*	Common adaptive cutoff is not map-ready
96% censored or unresolved	Do not fill with the discovery boundary
0 of 25 discovery-stable	More support changed the horizon inference

Experiment 2: characterize decay

10. NEW QUESTION

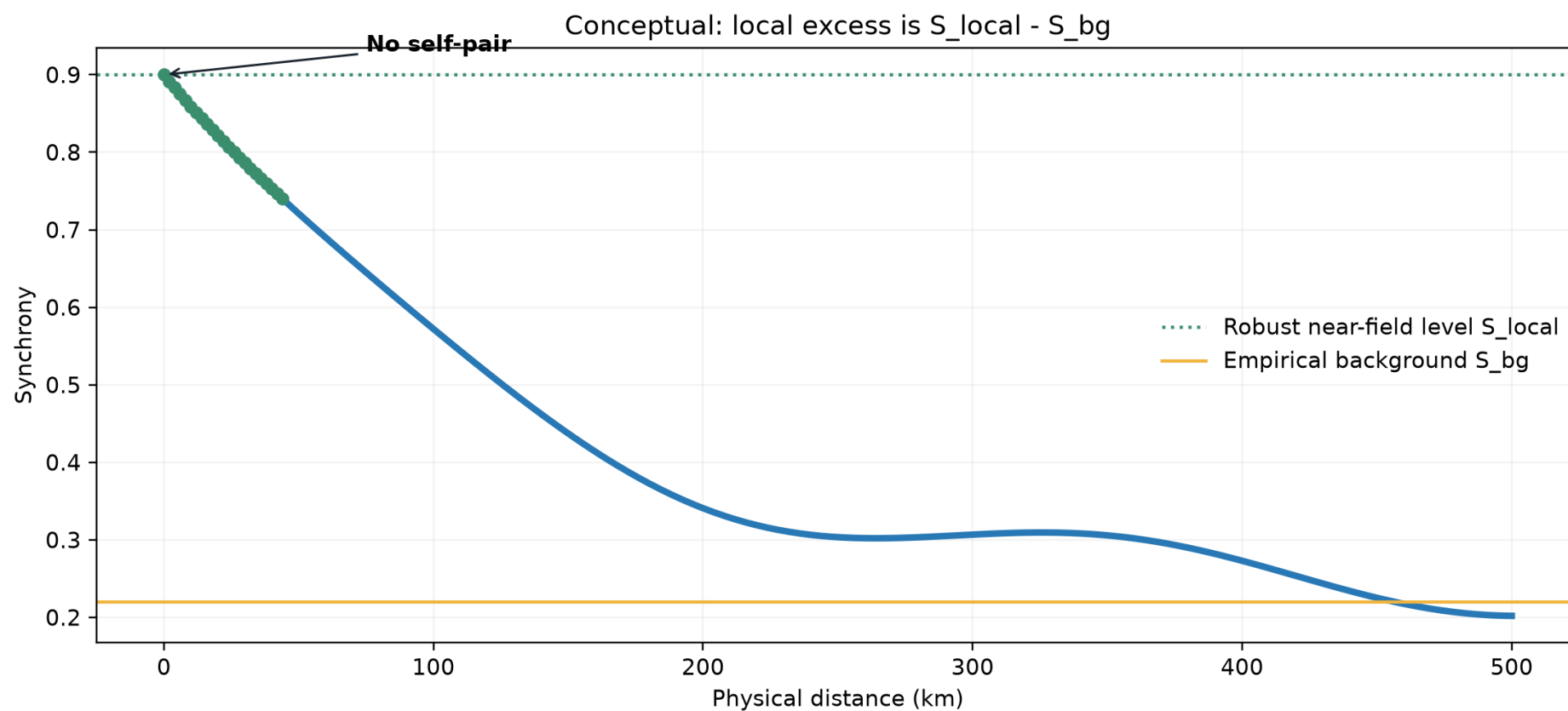
The same pair relationships can support characteristic curve properties even when they do not support a finite endpoint. Experiment 2 extracts $d_{25}/d_{50}/d_{75}$, effective length L , and robust local slope β .



Near-field synchrony

11. S_LOCAL

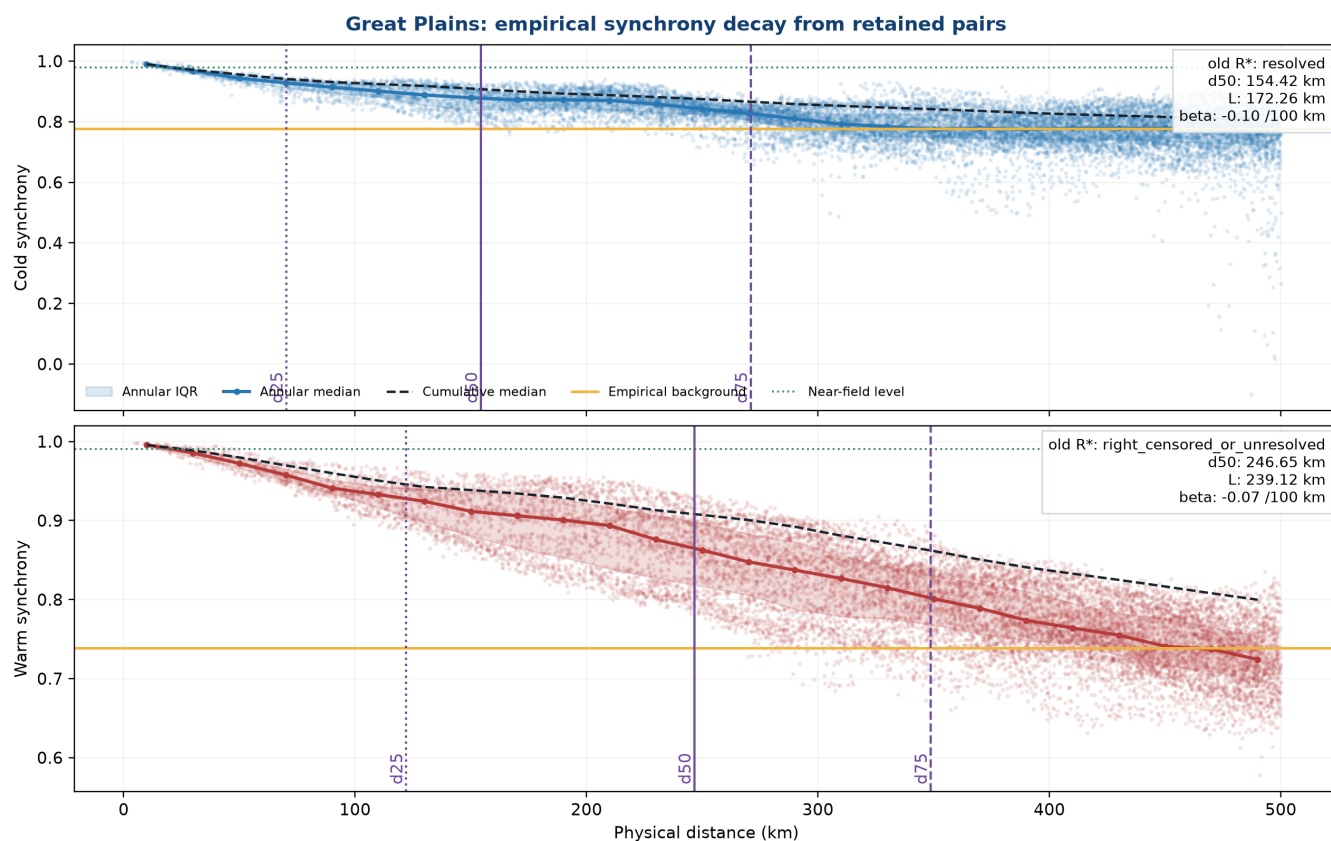
S_{local} is a robust median of the first two well-supported physical annuli. The self-pair remains excluded, so the near-field level is observed from neighboring cells rather than assumed to equal one.



Distant empirical background

12. S_BG

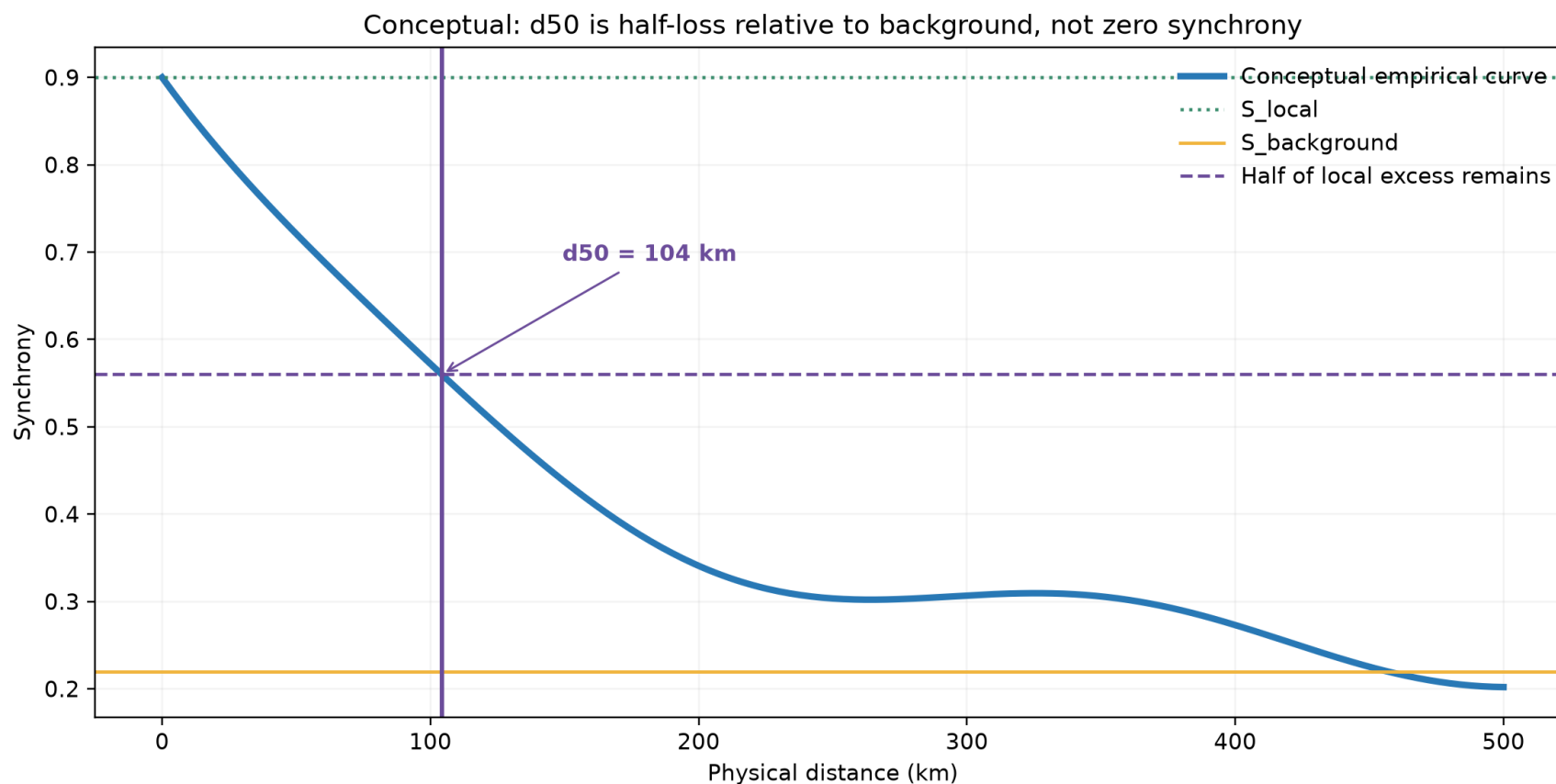
The existing outer-annulus, smoothed-outer, and distant-pair backgrounds are retained. Background sensitivity is measured explicitly instead of choosing whichever definition maximizes coverage.



Normalize local excess synchrony

13. E(D)

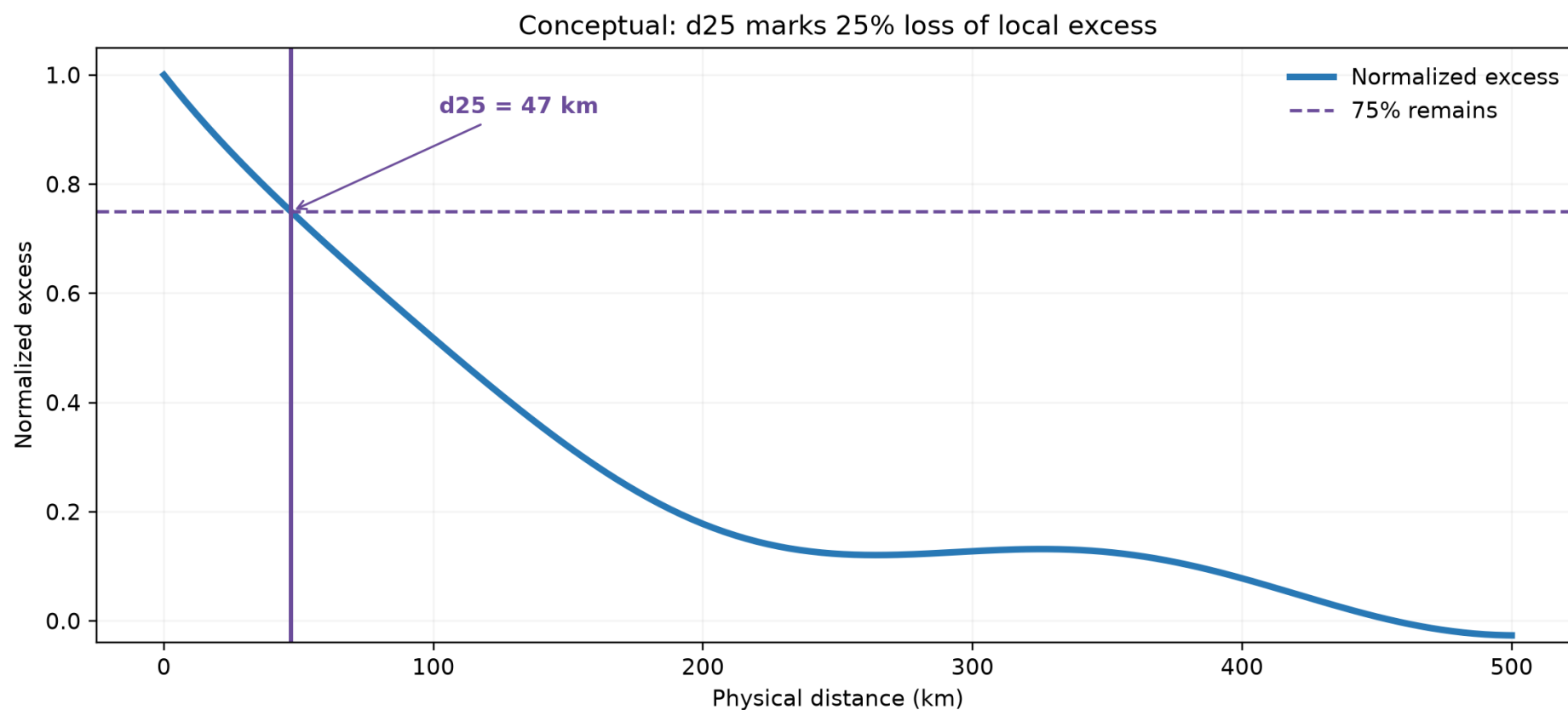
Normalized excess is $[S(d) - S_{bg}] / [S_{local} - S_{bg}]$. It is approximately one near the focal pixel and approaches zero near background. Curves with insufficient or nonpositive local excess retain explicit status flags.



d25: one quarter of excess has decayed

14. FRACTIONAL DISTANCE

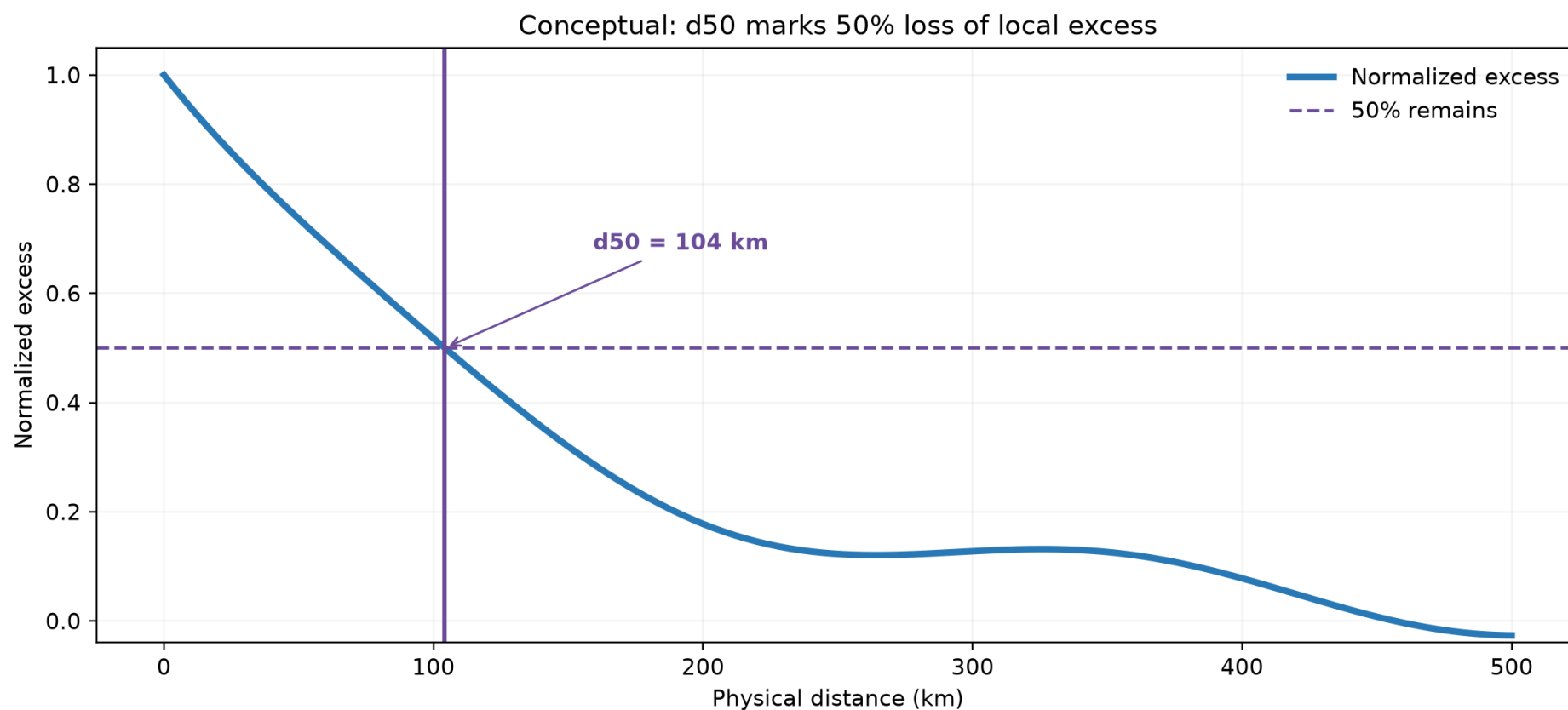
d25 is the interpolated physical distance where 75% of local excess remains. A centered three-annulus median and persistent later agreement prevent a single noisy bin from defining the crossing.



d50: half of local excess has decayed

15. PRIMARY FRACTIONAL DISTANCE

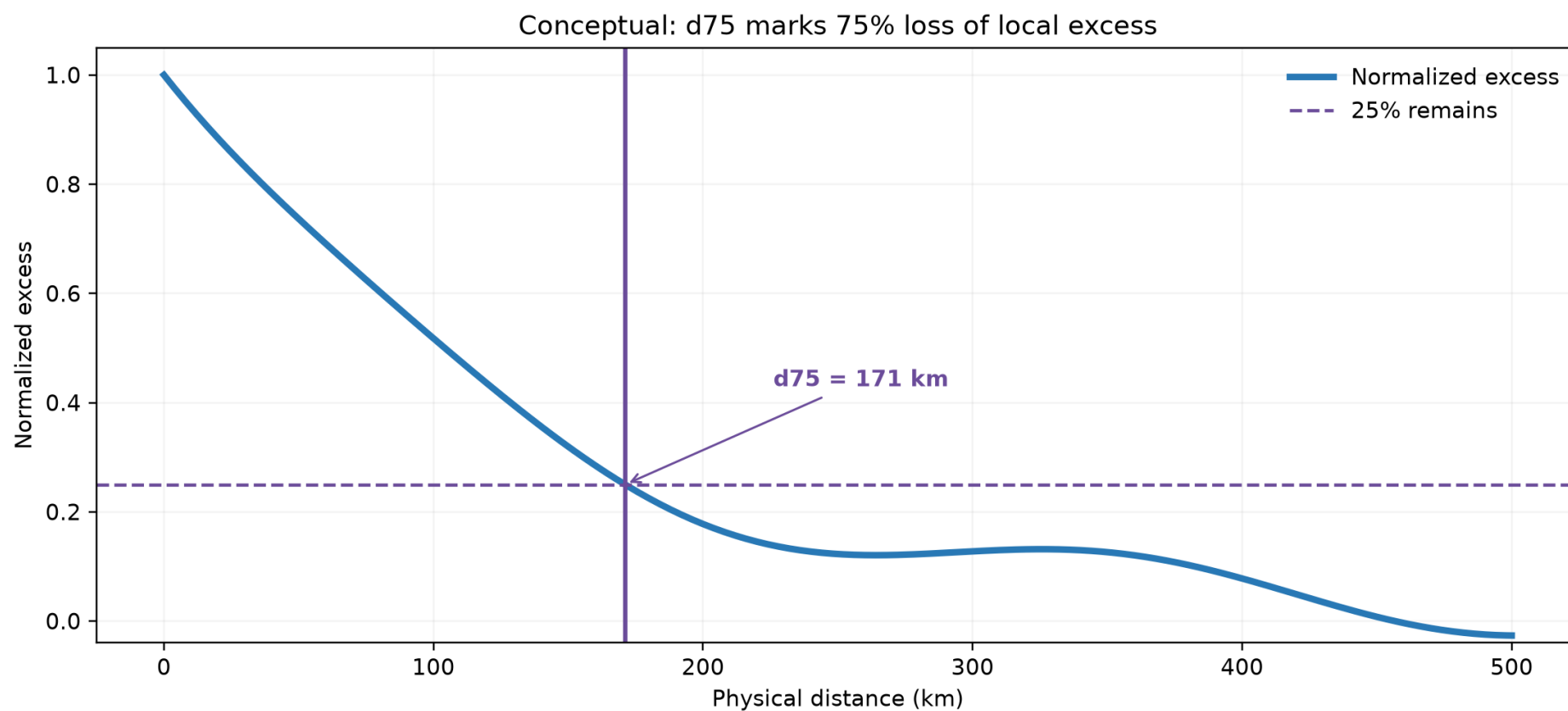
d50 is not zero synchrony and not disappearance. It is the distance where half of the locally elevated synchrony relative to the empirical background has been lost.



d75: three quarters of excess has decayed

16. FARTHER FRACTIONAL DISTANCE

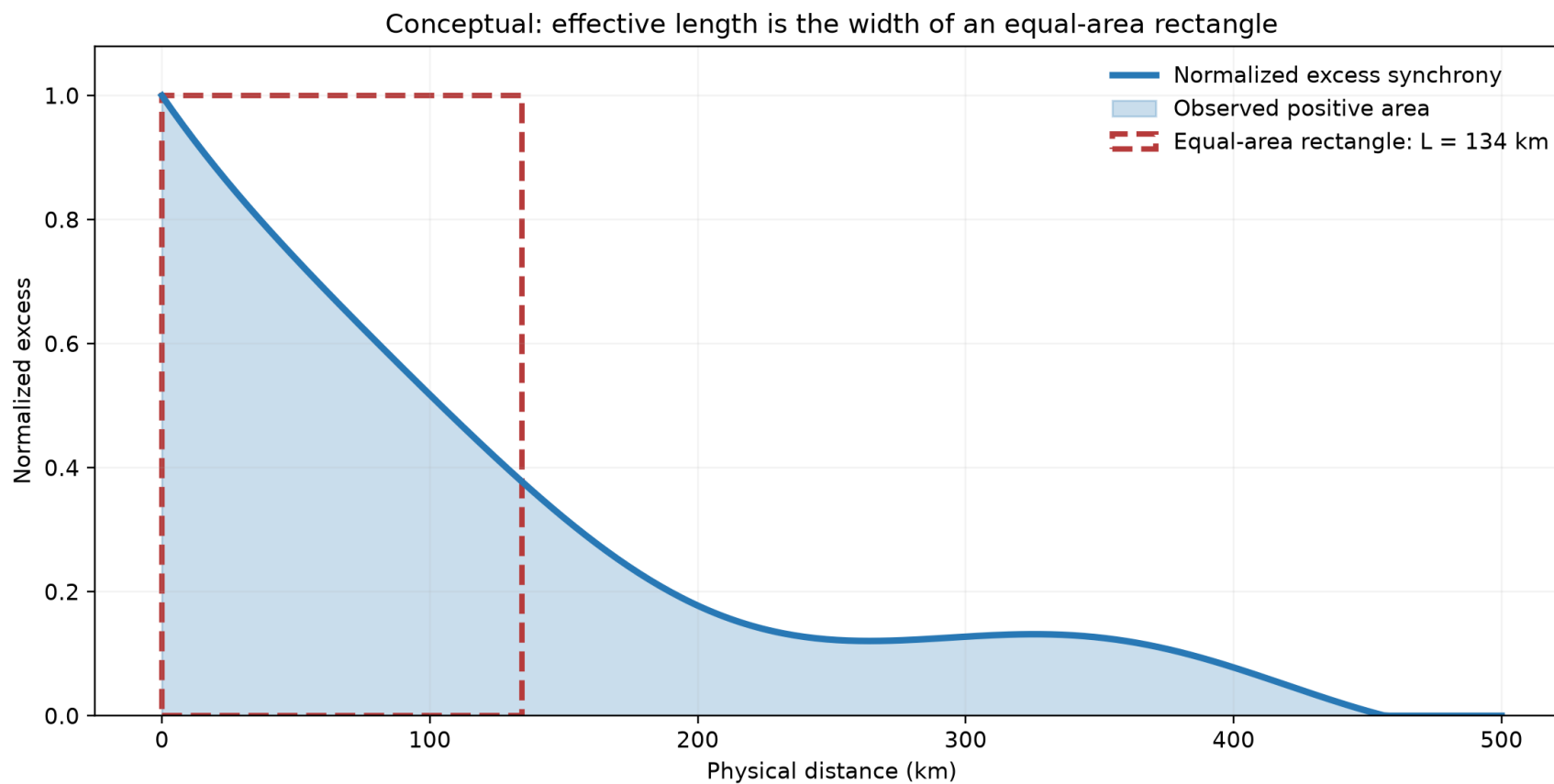
d75 targets 25% remaining excess and is expected to censor more often. Any fraction not reached within discovery support stays unresolved rather than being assigned 500 km.



Effective synchrony length L

17. INTEGRATED SCALE

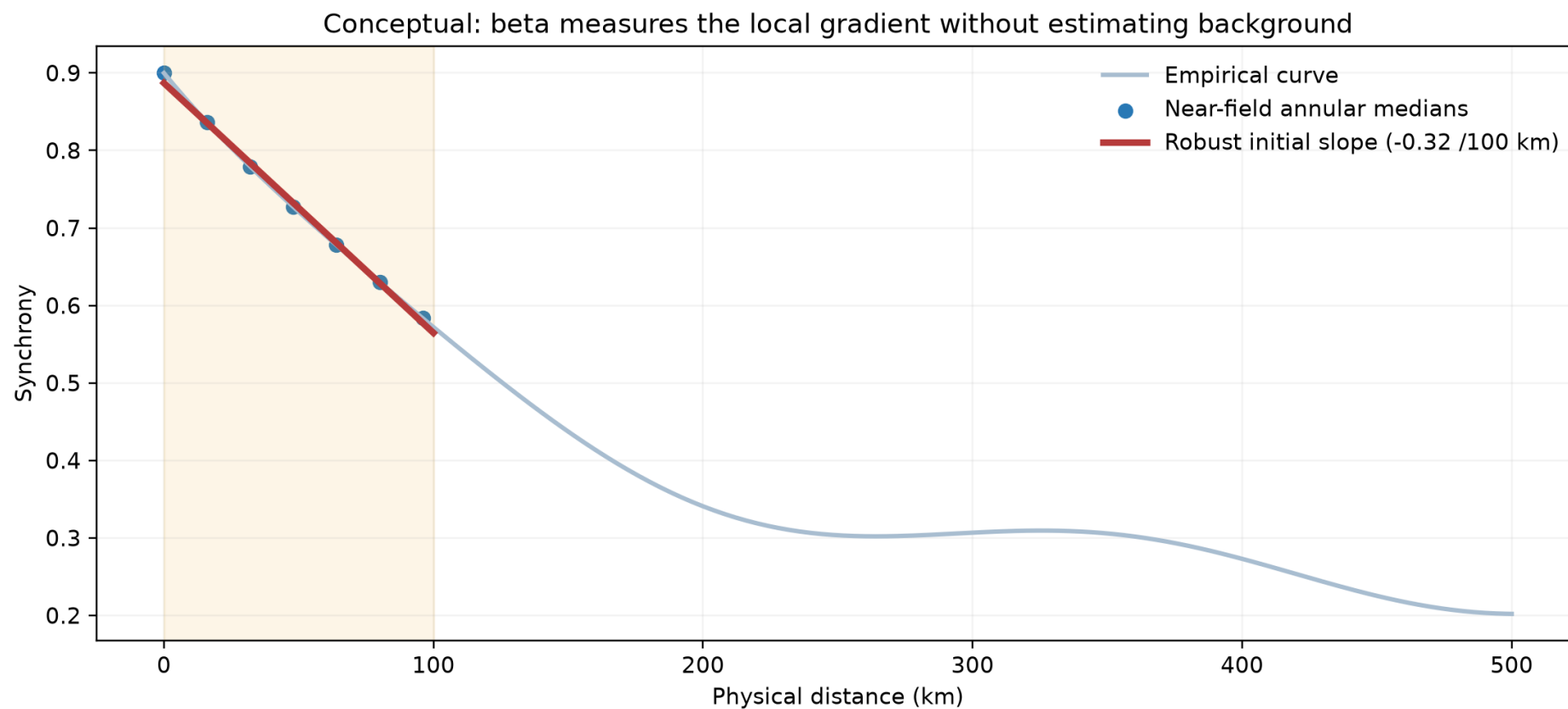
L is the positive area under normalized excess synchrony. It is the width of a height-one rectangle with equal area: a curve property in kilometers, not a hard cutoff, kernel bandwidth, or dispersal distance.



Initial decay slope beta

18. BACKGROUND-FREE SCALE

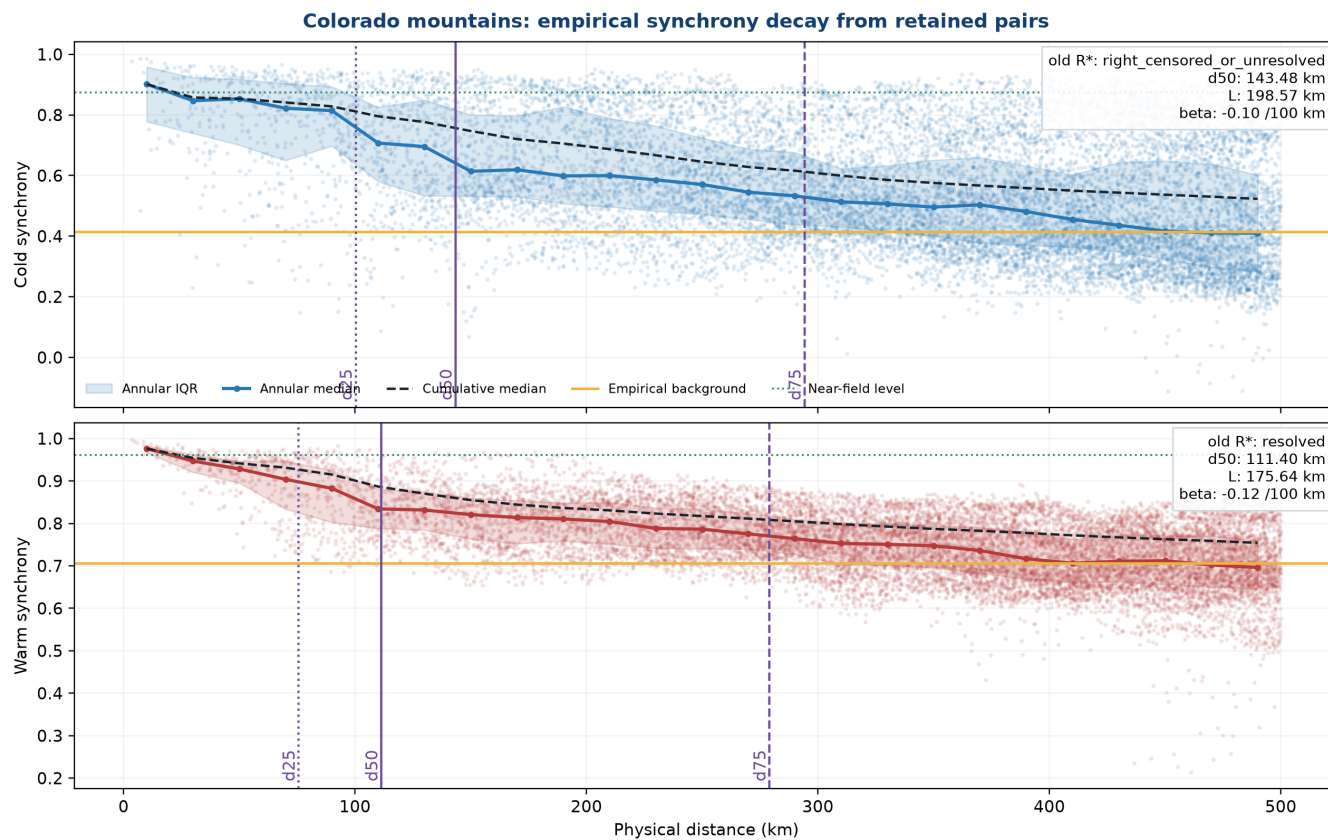
Beta is a Theil-Sen median slope from supported annular medians in the first 100 km, expressed per 100 km. It tests local decay even when estimating a distant background is difficult.



One real cold example

19. RETAINED PRISM PAIRS

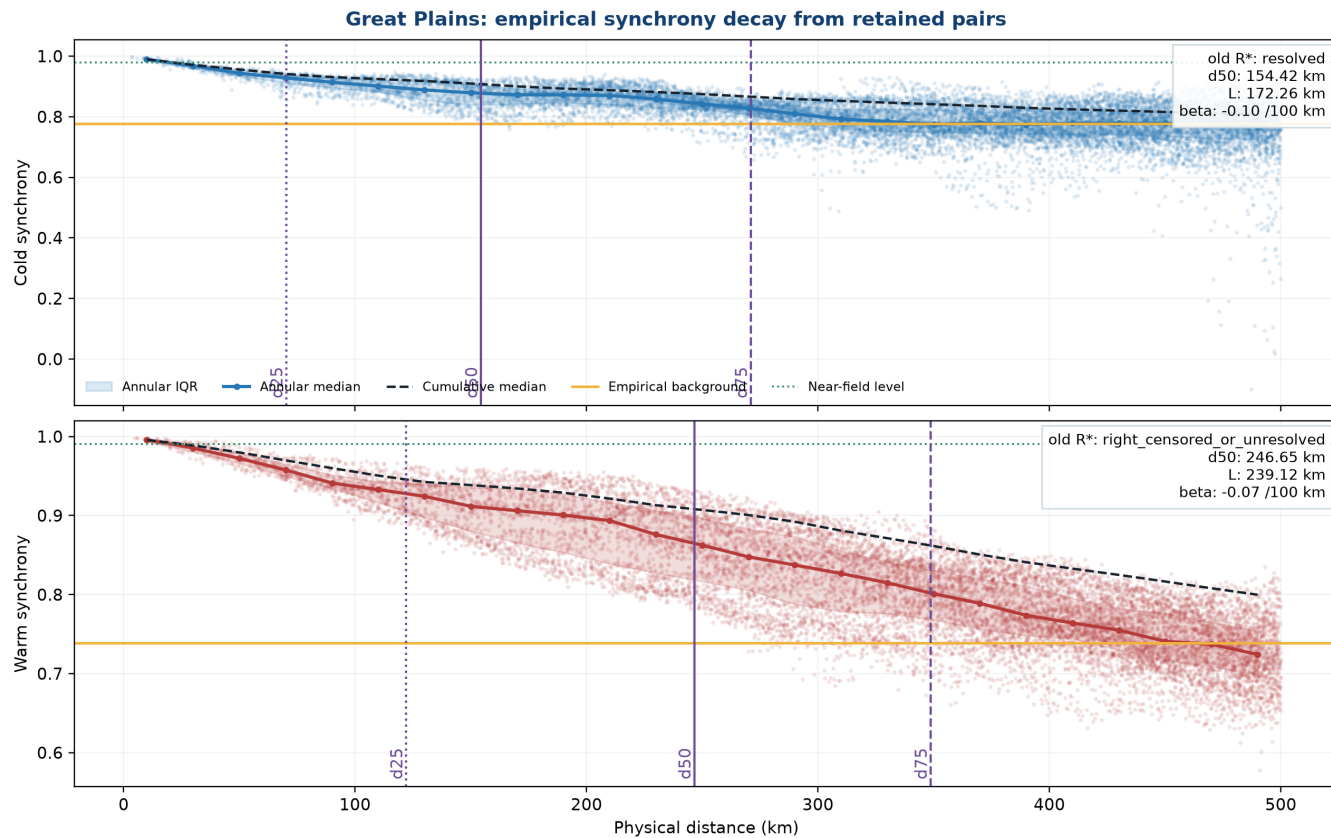
The Colorado mountain cold curve has a visible local decline and a long broad tail. The figure overlays fractional distances, effective length, beta, and the preserved old R* status on actual pair relationships.



One real warm example

20. RETAINED PRISM PAIRS

The Great Plains warm curve illustrates why a decay characteristic can resolve even when a stable finite horizon does not. The curve remains empirical and non-monotonic.

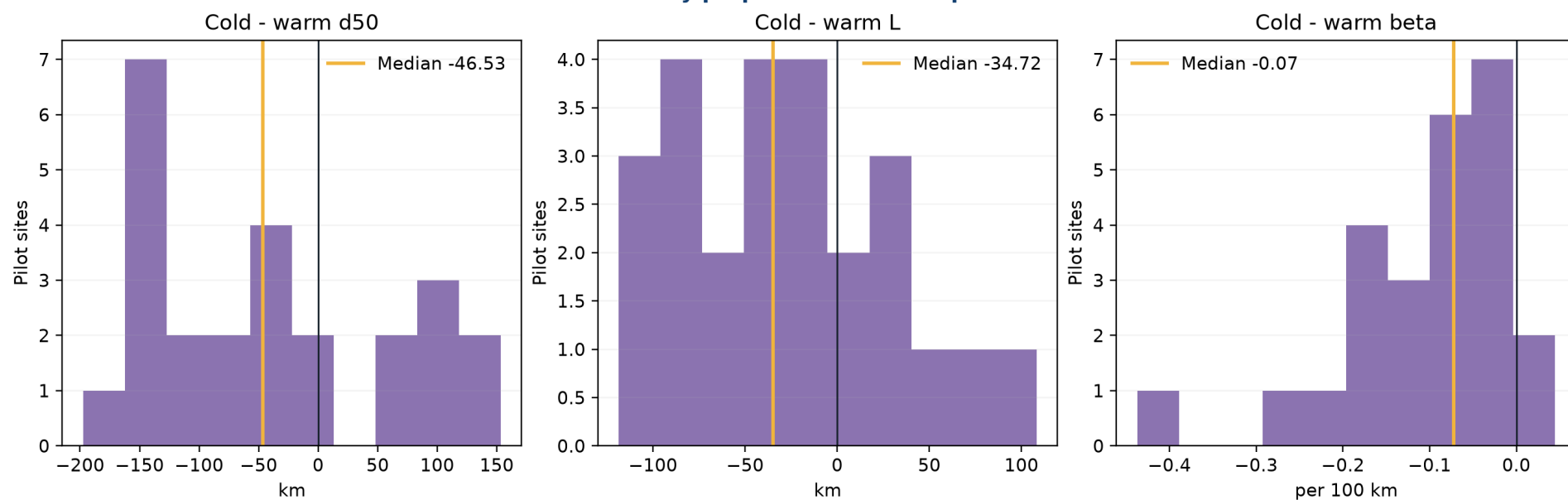


Cold and warm remain separate

21. PAIRED CONTRASTS

All metrics are estimated independently for lower-tail TMIN and strict upper-tail TMAX. Cold-minus-warm d50, L, and beta are distinct from Delta_S and are never merged into a synthetic index.

Cold and warm decay properties remain separate contrasts

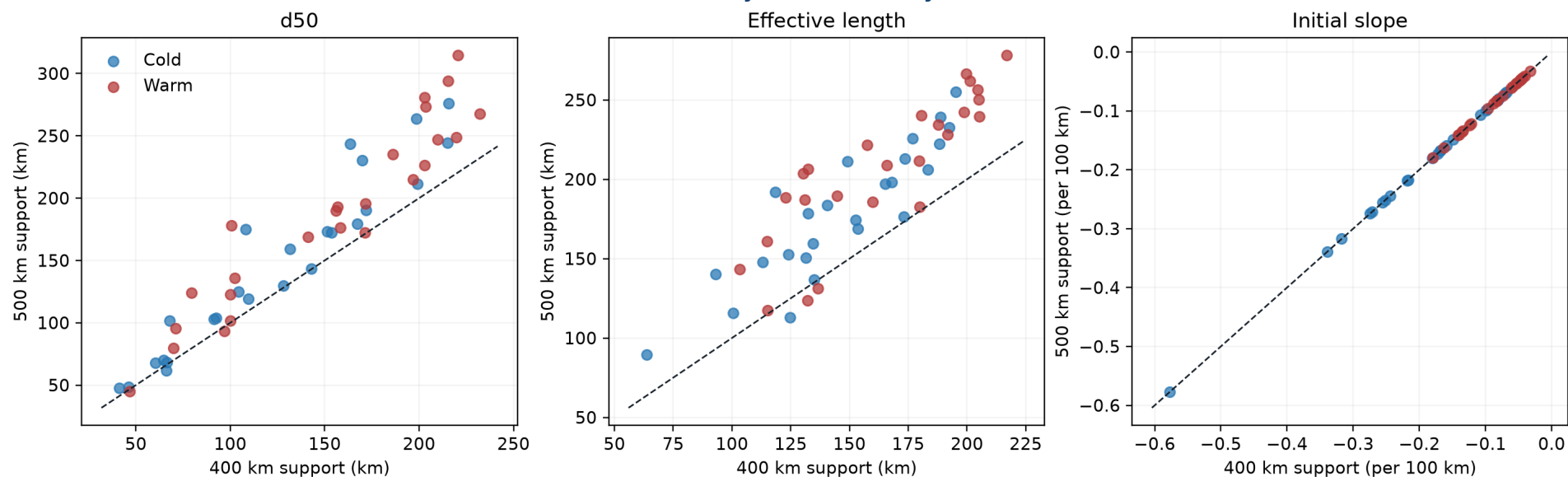


Sensitivity to 200/300/400/500 km

22. DISCOVERY SUPPORT

d50 is far more identifiable than R^* , but warm d50 is stable from 400 to 500 km at only 52% of Colorado sites. L stabilizes at only 12% of cold and 16% of warm sites. Beta is discovery-stable at all sites with support.

Discovery-radius stability

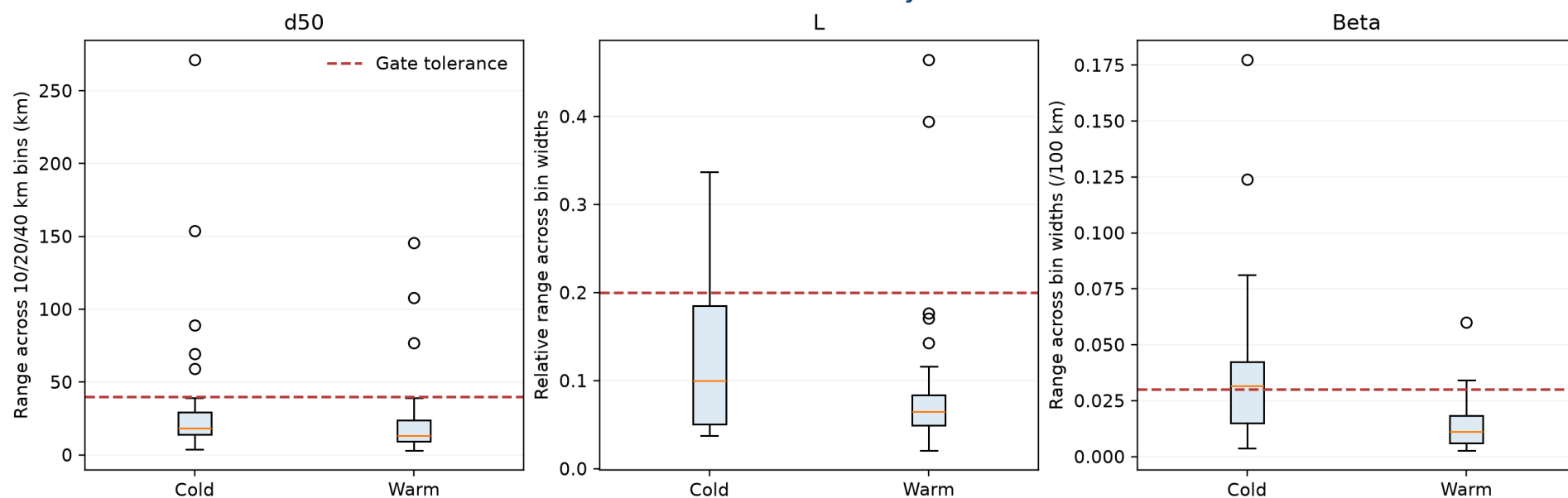


Sensitivity to 10/20/40 km bins

23. ANNULAR RESOLUTION

Cold and warm d50 meet the bin-width gate at 80% and 88% of sites. Effective length meets it at 76% and 92%. Cold beta meets it at only 40%, preventing beta from passing overall.

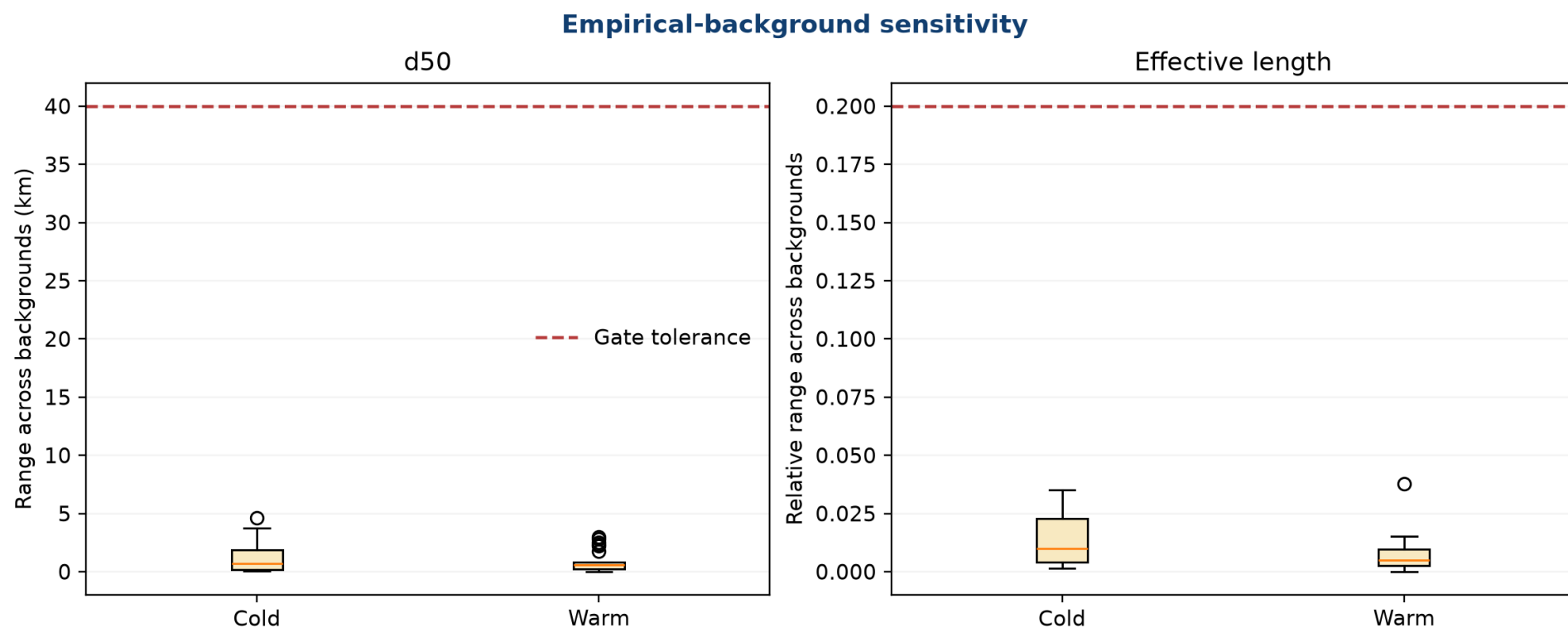
Bin-width sensitivity



Sensitivity to empirical background

24. BACKGROUND DEFINITION

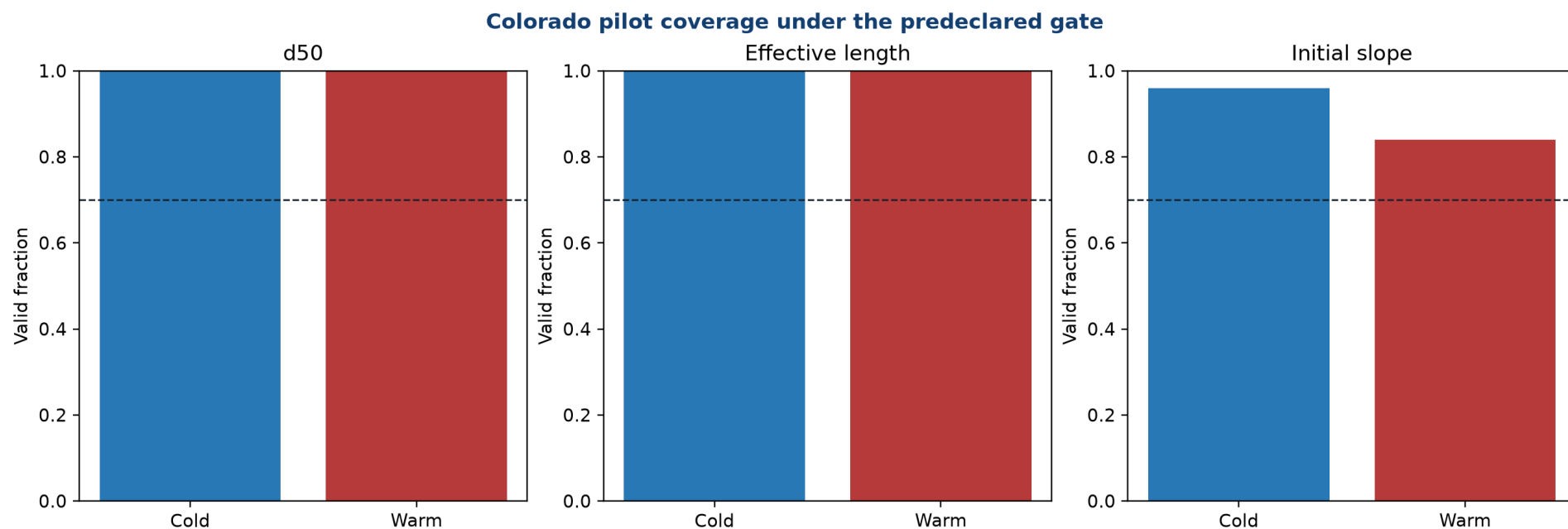
d50 and L meet their background-sensitivity tolerances at all 25 sites under the three retained definitions. Beta is exactly background-independent by construction.



Which metrics resolve?

25. COVERAGE

At 500 km / 20 km, d50 resolves for cold and warm at all 25 Colorado sites and all five representative sites. L is calculable at all sites. Beta meets its fit-quality requirement at 96% of cold and 84% of warm sites.



Which metrics stabilize?

26. GATE COMPONENTS

Coverage is not the decision. d50 fails on warm discovery stability; L fails strongly on discovery-boundary dependence; beta fails on cold bin-width sensitivity.

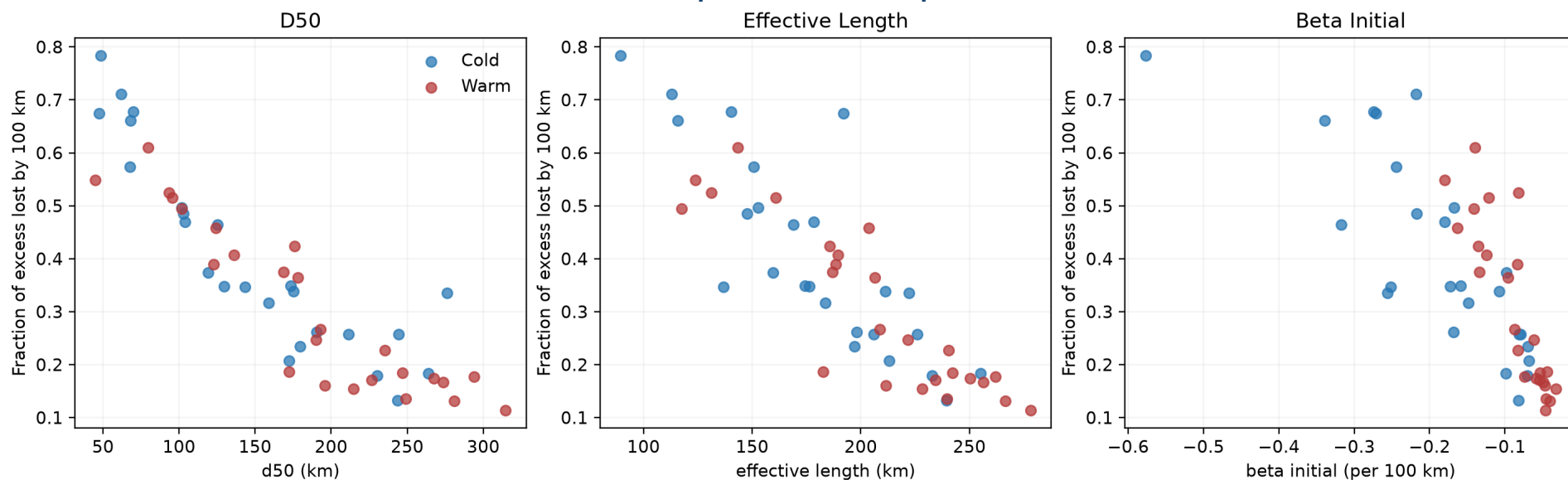
Metric	Cold	Warm
d50 discovery stable	76%	52%
d50 bin stable	80%	88%
L discovery stable	12%	16%
Beta discovery stable	100%	100%
Beta bin stable	40%	84%

What does 100 km represent?

27. FIXED-RADIUS INTERPRETATION

The median fraction of local excess lost by 100 km is 0.35 for cold and 0.25 for warm. The fixed radius samples meaningful local decay, not a universal half-decay or endpoint.

What 100 km represents on the empirical curves

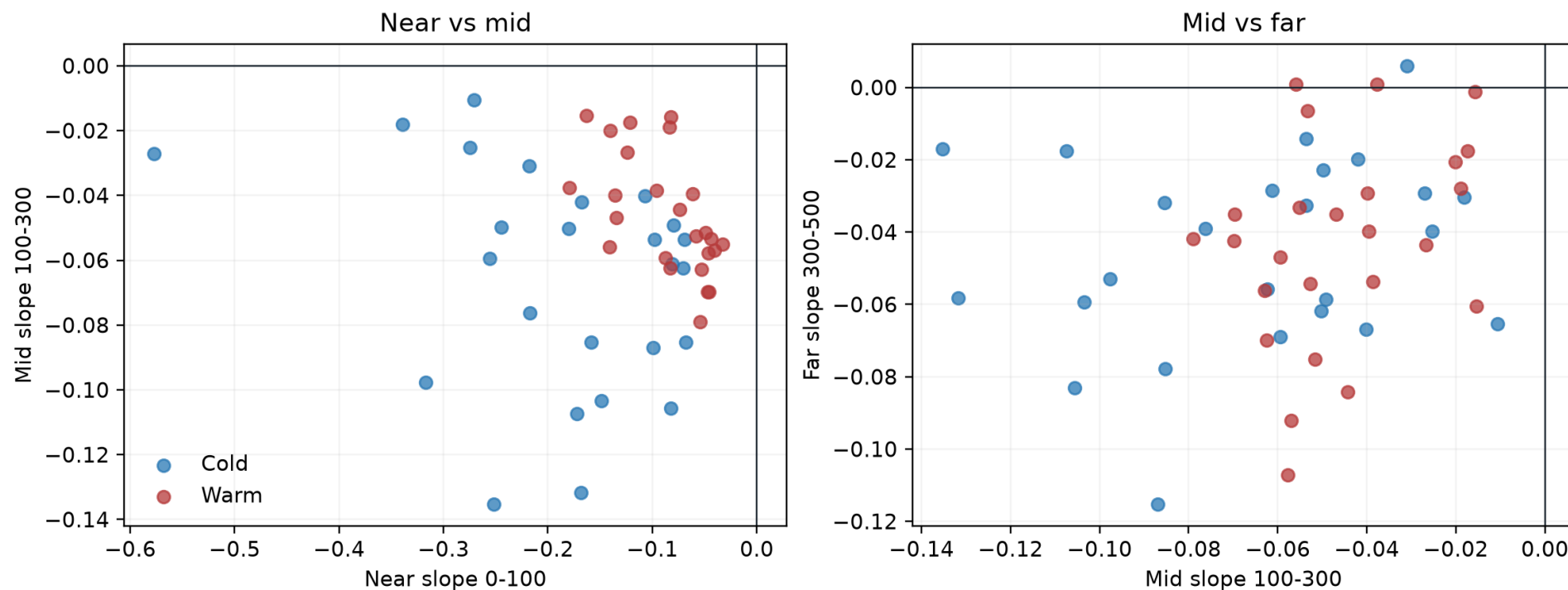


Evidence for multiple spatial scales

28. NEAR, MIDDLE, AND FAR SLOPES

Near-field decline is steeper than the far-field tail at 84% of cold sites and 56% of warm sites. This supports a fast local component plus a broad slower component, explaining why 100 km can be useful while R^* remains unresolved.

Multiscale decay diagnostic



Predeclared pilot decision gate

29. PASS/HOLD

A metric must pass coverage, representative-site, discovery, bin-width, and background criteria for both tails. None passes the complete gate; no statewide raster is authorized by this experiment.

Candidate	Decision and limiting evidence
d50	HOLD: warm discovery stability 52% < 70%
Effective length L	HOLD: discovery stability 12% cold / 16% warm
Initial beta	HOLD: cold bin stability 40% < 70%
Full Colorado	HOLD
CONUS	HOLD

Recommended next experiment

30. DECISION AND NEXT STEP

The proposition is partly supported: measurable decay properties exist without a sharp horizon, but none is ready for mapping under the complete gate. Next, test warm d50 and cold beta on independent seasonal windows and evaluate a bin-width-robust local-gradient summary. Do not tune thresholds merely to obtain PASS. Reproduce with:

```
.venv/bin/python scripts/run_empirical_synchrony_decay_pilot.py
```

Question	Answer
Does d50 resolve more often than R*?	Yes: 25/25 per tail and 5/5 representatives versus 1/25 common R*
Does L stabilize?	Usually no; strongly discovery-boundary dependent
Is beta stable?	Discovery-stable and background-free, but cold is bin-sensitive
Does 100 km remain useful?	Yes as a local-scale control, not as a horizon
Proceed to statewide or CONUS maps?	No - HOLD all candidates